Zewail City of Science and Technology
University of Science and Technology

Design and Performance Analysis of a Supersonic Ramjet Engine

SPC 410 - Gas Dynamics Project
Fall 2025

Team Members:

Asmaa Sapry 202201338
Raed Asham 202200272

Supervisors:

Dr. Ahmed T. Eltaweel
Eng. Asmaa Alaa ElDeen

January 21, 2026

Chapter 1

Introduction

Supersonic airbreathing propulsion systems provide an efficient means of producing thrust at high flight speeds by utilizing atmospheric oxygen rather than carrying an onboard oxidizer. Among these systems, the ramjet engine is one of the simplest configurations capable of efficient operation in the supersonic regime. A ramjet produces thrust by compressing the incoming freestream through shock waves, adding heat through combustion, and expanding the high–energy flow through a nozzle to generate momentum increase.

The objective of this project is to design and analyze a two–dimensional ramjet engine operating at a specified design–point flight condition. The engine consists of four main components: an external compression intake, an isentropic diffuser, a constant–area combustion chamber, and a convergent–divergent nozzle. Each component is modeled using one–dimensional gas–dynamic relations under standard simplifying assumptions, including steady flow, perfect gas behavior, and adiabatic walls except in the combustor where heat addition is present.

At the design point, the intake geometry is optimized to maximize total pressure recovery while maintaining attached oblique shocks. The diffuser is then designed to decelerate the subsonic flow to a low Mach number suitable for stable combustion. The combustor is analyzed using Rayleigh flow relations with a prescribed maximum total temperature and additional total pressure losses to account for friction and heat release. Finally, the nozzle is designed to expand the combustion products to ambient pressure to produce thrust.

In addition to the design–point analysis, a bonus off–design study is performed by evaluating the ramjet performance at a higher flight Mach number while keeping the engine geometry fixed. This provides insight into the sensitivity of engine performance to changes in flight conditions.

The results of this project include optimized intake ramp angles, geometric dimensions of the engine, flow properties at each major station, and overall engine performance parameters such as mass flow rate and net thrust.

Chapter 2

Intake Design and Optimization

(MatLab Code found at Appendix 9.1)

2.1 Objective

The purpose of the intake is to compress the incoming supersonic flow to a subsonic Mach number suitable for the diffuser and combustor, while maximizing total pressure recovery. A two–ramp external compression intake followed by a terminal normal shock is employed. The design objective is therefore to maximize:

$$\eta_{intake} = \frac{P_{0,4}}{P_{0,1}} \tag{2.1}$$

where station (1) denotes freestream conditions and station (4) denotes intake exit conditions.

2.2 Freestream Conditions

The design-point flight conditions specified in the project statement are:

$$M_1 = 2.75 \tag{2.2}$$
$$P_1 = 12.11 \text{ kPa} \tag{2.3}$$
$$T_1 = 216.5 \text{ K} \tag{2.4}$$

The corresponding total freestream conditions are obtained from isentropic relations:

$$T_{0,1} = T_1 \left(1 + \frac{\gamma - 1}{2} M_1^2\right) \tag{2.5}$$
$$P_{0,1} = P_1 \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{2.6}$$

with $\gamma = 1.4$ for air.

2.3 Oblique Shock Modeling

Each intake ramp generates an attached oblique shock. For a given upstream Mach number M and ramp deflection angle θ, the shock angle β is obtained from the θ–β–M relation:

$$\tan\theta = 2\cot\beta \frac{M^2\sin^2\beta - 1}{M^2(\gamma + \cos 2\beta) + 2} \tag{2.7}$$

The normal component of the upstream Mach number is:

$$M_{n1} = M\sin\beta \tag{2.8}$$

The total pressure loss across each shock is computed using the normal shock total pressure relation evaluated at M_{n1}. The downstream Mach number after the oblique shock is:

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)} \tag{2.9}$$

The same procedure is applied to the second ramp, using M_2 as the upstream Mach number.

2.4 Terminal Normal Shock

After the second oblique shock, the flow remains supersonic and a terminal normal shock is required to bring the flow to subsonic conditions. The downstream Mach number M_4 and total pressure loss are computed using standard normal shock relations.

The overall intake total pressure recovery is therefore:

$$\frac{P_{0,4}}{P_{0,1}} = \left(\frac{P_{0,2}}{P_{0,1}}\right)\left(\frac{P_{0,3}}{P_{0,2}}\right)\left(\frac{P_{0,4}}{P_{0,3}}\right) \tag{2.10}$$

2.5 Optimization Procedure

A parametric sweep over the two ramp deflection angles θ_1 and θ_2 was performed. For each candidate pair:

- Oblique shock relations were solved to obtain β_1 and β_2
- Downstream Mach numbers M_2 and M_3 were computed
- Feasibility checks ensured attached shocks
- A terminal normal shock was applied
- The total pressure recovery was evaluated

The optimal configuration corresponds to the maximum feasible total pressure recovery.

2.6 Optimization Results

The optimization evaluated 54,440 feasible intake configurations. The optimal solution obtained is:

$$\theta_1 = 14.30^\circ, \quad \theta_2 = 17.40^\circ \tag{2.11}$$
$$\beta_1 = 33.58^\circ, \quad \beta_2 = 46.31^\circ \tag{2.12}$$
$$M_2 = 2.1013 \tag{2.13}$$
$$M_3 = 1.4360 \tag{2.14}$$
$$M_4 = 0.7250 \tag{2.15}$$

The resulting total pressure recovery is:

$$\frac{P_{0,4}}{P_{0,1}} = 0.808695 \tag{2.16}$$

Thus, approximately 81% of the freestream stagnation pressure is preserved through the intake.

2.7 Intake Geometry

The engine internal duct has a square cross–section of side length $h = 0.25$ m. The wedge lengths are estimated geometrically as:

$$L_1 = \frac{h}{\tan\beta_1}, \qquad L_2 = \frac{h}{\tan\beta_2} \tag{2.17}$$

which yield:

$$L_1 = 0.3766 \text{ m} \tag{2.18}$$
$$L_2 = 0.2388 \text{ m} \tag{2.19}$$

The intake exit area (station 4), which serves as the diffuser inlet, is:

$$A_4 = 0.02272 \text{ m}^2 \tag{2.20}$$

2.8 Optimization Map

Figure 2.1 shows the contour map of total pressure recovery as a function of the two ramp angles. The white marker denotes the optimal design point.

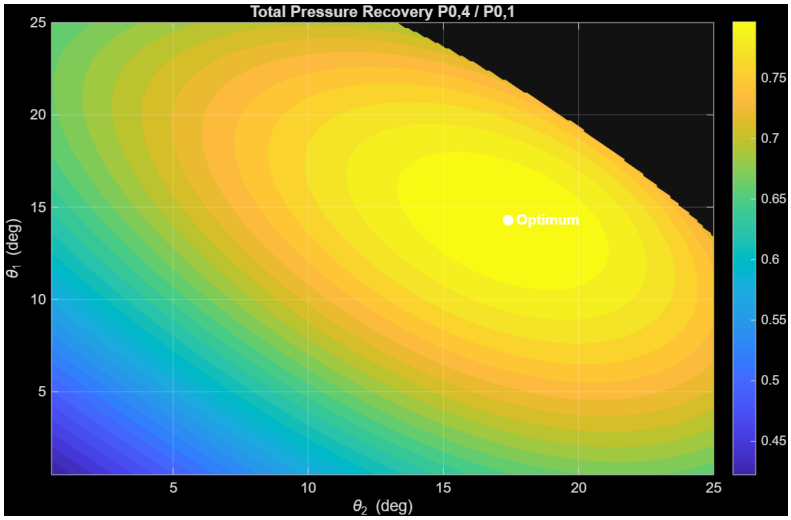


Figure 2.1: Contour map of total pressure recovery $P_{0,4}/P_{0,1}$ as a function of ramp angles θ_1 and θ_2.

2.9 MATLAB Optimization Output

For completeness, the MATLAB optimization output confirming the selected design is shown in Figure 2.2.

```
================ OPTIMUM INTAKE DESIGN ================
Feasible designs checked: 54440

Best theta1  = 14.300 deg
Best theta2  = 17.400 deg
Best beta1   = 33.579 deg
Best beta2   = 46.314 deg
Mach after shock 1 (M2) = 2.1013
Mach after shock 2 (M3) = 1.4360
Mach after normal shock (M4) = 0.7250
Total pressure recovery (P0,4/P0,1) = 0.808695
=======================================================
```

Figure 2.2: MATLAB command window output for intake optimization results.

2.10 Summary

The optimized two–ramp external compression intake compresses the incoming freestream from $M_1 = 2.75$ to a subsonic exit Mach number of $M_4 = 0.725$, while achieving a total pressure recovery of $P_{0,4}/P_{0,1} = 0.8087$. This provides suitable inlet conditions for the diffuser and subsequent combustion process.

Chapter 3

Diffuser Design

(MatLab Code found at Appendix 9.2)

3.1 Objective

Following the intake, the flow must be further decelerated to a low subsonic Mach number before entering the combustion chamber. This is achieved using a quasi–one–dimensional isentropic diffuser. The diffuser is designed to reduce the Mach number while increasing static pressure, without incurring additional total pressure losses.

3.2 Inlet Conditions

The diffuser inlet corresponds to the intake exit (station 4). From the intake optimization results:

$$M_4 = 0.725 \tag{3.1}$$
$$P_{0,4} = 246206.86 \text{ Pa} \tag{3.2}$$
$$T_{0,4} = 543.96 \text{ K} \tag{3.3}$$

Since the diffuser is assumed adiabatic and frictionless, the total conditions remain constant throughout the diffuser:

$$P_{0,5} = P_{0,4}, \qquad T_{0,5} = T_{0,4} \tag{3.4}$$

3.3 Isentropic Diffuser Relations

The static pressure and temperature at any section are obtained from isentropic relations:

$$T = \frac{T_0}{1 + \frac{\gamma-1}{2}M^2} \tag{3.5}$$
$$P = \frac{P_0}{\left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}} \tag{3.6}$$

To achieve efficient combustion, a low subsonic exit Mach number is desired. A target diffuser exit Mach number of:

$$M_5 = 0.20 \tag{3.7}$$

was selected.

The required diffuser area ratio is then obtained from the quasi–one–dimensional area–Mach relation:

$$\frac{A_5}{A_4} = \frac{M_4}{M_5}\left(\frac{1+\frac{\gamma-1}{2}M_5^2}{1+\frac{\gamma-1}{2}M_4^2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{3.8}$$

3.4 Diffuser Results

Applying the above relations yields the following diffuser performance:

$$M_4 = 0.7250 \tag{3.9}$$
$$M_5 = 0.2000 \tag{3.10}$$

Total conditions (constant through diffuser):

$$P_{0,4} = P_{0,5} = 246206.86 \text{ Pa} \tag{3.11}$$
$$T_{0,4} = T_{0,5} = 543.96 \text{ K} \tag{3.12}$$

Static conditions:

$$P_4 = 173524.27 \text{ Pa}, \quad T_4 = 492.21 \text{ K} \tag{3.13}$$
$$P_5 = 239435.36 \text{ Pa}, \quad T_5 = 539.64 \text{ K} \tag{3.14}$$

Required diffuser area ratio:

$$\frac{A_5}{A_4} = 2.7508 \tag{3.15}$$

Thus, the diffuser successfully decelerates the flow from $M_4 = 0.725$ to $M_5 = 0.20$, while increasing static pressure and temperature as expected for an isentropic diffusion process.

3.5 MATLAB Diffuser Output

Figure 3.1 shows the MATLAB output used to compute the diffuser properties.

```
================ DIFFUSER RESULTS ================
Inlet Mach (after intake)      M4 = 0.7250
Exit Mach (combustor entry)    M5 = 0.2000

Total Pressure (constant)      P04 = 246206.86 Pa
Total Temperature (constant)   T04 = 543.96 K

Static Pressure at diffuser inlet  P4 = 173524.27 Pa
Static Pressure at diffuser exit   P5 = 239435.36 Pa

Static Temperature at inlet  T4 = 492.21 K
Static Temperature at exit   T5 = 539.64 K

Required Area Ratio A5/A4 = 2.7508
==================================================
```

Figure 3.1: MATLAB command window output for diffuser calculations.

3.6 Summary

The diffuser reduces the intake exit Mach number from $M_4 = 0.725$ to a low subsonic Mach number of $M_5 = 0.20$, providing suitable inlet conditions for the combustion chamber. The required diffuser area ratio is $A_5/A_4 = 2.7508$, ensuring efficient static pressure recovery prior to heat addition in the combustor.

Chapter 4

Combustion Chamber Design

(MatLab Code found at Appendix 9.3)

4.1 Objective

After diffusion, heat is added to the flow in the combustion chamber to increase the total temperature of the gas and provide the energy necessary for thrust generation. The combustor is modeled as a one–dimensional constant–area duct with heat addition and is analyzed using Rayleigh flow relations. In addition, a 5% total pressure loss is included to account for frictional and non-ideal effects.

4.2 Inlet Conditions

The combustor inlet corresponds to the diffuser exit (station 5). From the diffuser calculations:

$$M_5 = 0.2000 \tag{4.1}$$
$$P_{0,5} = 246206.86 \text{ Pa} \tag{4.2}$$
$$T_{0,5} = 543.96 \text{ K} \tag{4.3}$$
$$P_5 = 239435.36 \text{ Pa} \tag{4.4}$$
$$T_5 = 539.64 \text{ K} \tag{4.5}$$

4.3 Rayleigh Flow Relations

For one–dimensional Rayleigh flow with heat addition in a constant–area duct, the governing relation between the local Mach number and total temperature is:

$$\frac{T_0}{T_0^*} = \frac{(\gamma+1)^2 M^2}{(1+\gamma M^2)^2} \frac{1+\frac{\gamma-1}{2}M^2}{1+\frac{\gamma-1}{2}} \tag{4.6}$$

where T_0^* denotes the critical total temperature at which the flow becomes choked ($M = 1$) on the Rayleigh line.

The maximum allowable combustor total temperature is specified as:

$$T_{0,6,\max} = 2200 \text{ K} \tag{4.7}$$

If the Rayleigh choking limit is reached before this value, the flow becomes choked in the combustor. In the present design, the maximum total temperature coincides with the Rayleigh limit.

4.4 Total Pressure Loss

The total pressure change due to Rayleigh heat addition is obtained from the Rayleigh total pressure ratio:

$$\frac{P_{0,6}}{P_{0,5}} = \frac{(P_0/P_0^*)_6}{(P_0/P_0^*)_5} \tag{4.8}$$

To account for additional frictional and mixing losses, a further 5% total pressure reduction is applied:

$$P_{0,6,\text{final}} = 0.95\, P_{0,6,\text{Rayleigh}} \tag{4.9}$$

4.5 Combustor Results

Solving the Rayleigh flow relations yields the following combustor performance:

$$M_6 = 0.5073 \tag{4.10}$$
$$T_{0,6} = 2200.00 \text{ K} \tag{4.11}$$
$$P_{0,6,\text{Rayleigh}} = 221570.30 \text{ Pa} \tag{4.12}$$
$$P_{0,6,\text{final}} = 210491.78 \text{ Pa} \tag{4.13}$$

Static exit conditions:

$$T_6 = 2092.30 \text{ K} \tag{4.14}$$
$$P_6 = 185876.99 \text{ Pa} \tag{4.15}$$

The heat added per unit mass is:

$$q = c_p(T_{0,6} - T_{0,5}) = 1663.49 \text{ kJ/kg} \tag{4.16}$$

4.6 MATLAB Combustor Output

Figure 4.1 shows the MATLAB output confirming the Rayleigh flow calculations in the combustor.

```
================ COMBUSTOR RESULTS (RAYLEIGH) ================
Combustor entry (station 5):
  M5  = 0.2000
  T05 = 543.96 K (total)
  P05 = 246206.86 Pa (total)
  T5  = 539.64 K (static)
  P5  = 239435.36 Pa (static)

Combustor exit (station 6):
  M6  = 0.5073
  T06 = 2200.00 K (total)  [limit = 2200.00 K]
  P06 (Rayleigh only) = 221570.30 Pa (total)
  P06 (with extra 5% loss) = 210491.78 Pa (total)
  T6  = 2092.30 K (static)
  P6  = 185876.99 Pa (static)

Heat added per unit mass:
  q = cp*(T06 - T05) = 1663.49 kJ/kg
==============================================================
```

Figure 4.1: MATLAB output for Rayleigh-flow combustor calculations.

4.7 Summary

The combustion chamber increases the total temperature of the flow from $T_{0,5} = 543.96$ K to $T_{0,6} = 2200$ K through controlled heat addition. The exit Mach number rises to $M_6 = 0.5073$, while the total pressure is reduced to $P_{0,6} = 210491.78$ Pa due to Rayleigh heat addition and the prescribed 5% additional losses. The resulting conditions provide high-energy gas suitable for efficient expansion in the nozzle.

Chapter 5

Nozzle Design

(MatLab Code found at Appendix 9.4)

5.1 Objective

The final component of the ramjet engine is the convergent–divergent (C–D) nozzle. Its purpose is to expand the high–energy combustion products to ambient pressure and convert thermal energy into kinetic energy, thereby producing thrust. The nozzle is modeled as a quasi–one–dimensional isentropic flow passage.

5.2 Nozzle Inlet Conditions

The nozzle inlet corresponds to the combustor exit (station 6). From the combustor calculations:

$$T_{0,6} = 2200.00 \text{ K} \tag{5.1}$$
$$P_{0,6} = 210491.78 \text{ Pa} \tag{5.2}$$

The ambient (freestream) conditions are:

$$P_a = 12.11 \text{ kPa} \tag{5.3}$$
$$T_a = 216.5 \text{ K} \tag{5.4}$$

5.3 Isentropic Nozzle Relations

For isentropic expansion through the nozzle, the relation between total and static pressure is:

$$\frac{P_0}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{5.5}$$

The exit Mach number is obtained by solving:

$$P_e = \frac{P_{0,6}}{\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}} = P_a \tag{5.6}$$

The area–Mach relation for a quasi–one–dimensional nozzle is:

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{5.7}$$

Thus, the nozzle area ratio is:

$$\frac{A_e}{A_t} = \frac{A(M_e)}{A(M=1)} \tag{5.8}$$

5.4 Nozzle Results

Solving the isentropic nozzle relations yields:

$$M_e = 2.5110 \tag{5.9}$$
$$P_e = 12110.00 \text{ Pa} \tag{5.10}$$
$$T_e = 973.00 \text{ K} \tag{5.11}$$

The exit velocity is:

$$V_e = M_e\sqrt{\gamma R T_e} = 1570.05 \text{ m/s} \tag{5.12}$$

The nozzle geometric area ratio is:

$$\frac{A_e}{A_t} = 2.6640 \tag{5.13}$$

5.5 Mass Flow Rate and Thrust

The mass flow rate entering the engine is computed from the freestream capture area:

$$\dot{m} = \rho_1 V_1 A_{inlet} \tag{5.14}$$

The net thrust is obtained from the momentum equation:

$$F = \dot{m}(V_e - V_1) + (P_e - P_a)A_e \tag{5.15}$$

Since the nozzle is designed to expand the flow exactly to ambient pressure ($P_e = P_a$), the pressure thrust term vanishes.

The resulting performance parameters are:

$$\dot{m} = 9.880 \text{ kg/s} \tag{5.16}$$
$$F = 7498.45 \text{ N} \tag{5.17}$$

5.6 MATLAB Nozzle Output

Figure 5.1 shows the MATLAB output confirming the nozzle calculations.

```
================ NOZZLE RESULTS ================
Exit Mach number Me      = 2.5110
Exit static pressure Pe  = 12110.00 Pa
Exit static temperature  = 973.00 K
Exit velocity Ve         = 1570.05 m/s

Nozzle area ratio Ae/At  = 2.6640
Mass flow rate m_dot     = 9.880 kg/s
Net Thrust               = 7498.45 N
================================================
```

Figure 5.1: MATLAB output for convergent–divergent nozzle calculations.

5.7 Summary

The convergent–divergent nozzle expands the combustor exit flow from a total temperature of 2200 K and total pressure of 210 kPa to ambient pressure. The exit Mach number reaches $M_e = 2.511$, producing an exit velocity of 1570 m/s. With a nozzle area ratio $A_e/A_t = 2.664$, the engine produces a net thrust of approximately 7.5 kN at the design point.

Chapter 6

Overall Engine Results and Performance

(MatLab Code found at Appendix 9.5)

6.1 Final Engine Geometry

After completing the design of all engine components, the final geometric parameters of the ramjet are summarized below.

$$\theta_1 = 14.30^\circ, \quad \theta_2 = 17.40^\circ \tag{6.1}$$
$$\beta_1 = 33.58^\circ, \quad \beta_2 = 46.31^\circ \tag{6.2}$$

Estimated wedge lengths:

$$L_1 = 0.3766 \text{ m} \tag{6.3}$$
$$L_2 = 0.2388 \text{ m} \tag{6.4}$$

Cross–sectional areas at major stations:

$$A_4 = 0.022720 \text{ m}^2 \quad \text{(Intake exit / Diffuser inlet)} \tag{6.5}$$
$$A_5 = 0.062500 \text{ m}^2 \quad \text{(Diffuser exit / Combustor inlet)} \tag{6.6}$$
$$A_6 = 0.062500 \text{ m}^2 \quad \text{(Combustor exit)} \tag{6.7}$$
$$A_t = 0.054469 \text{ m}^2 \quad \text{(Nozzle throat)} \tag{6.8}$$
$$A_e = 0.145106 \text{ m}^2 \quad \text{(Nozzle exit)} \tag{6.9}$$

Nozzle geometric expansion ratio:

$$\frac{A_e}{A_t} = 2.6640 \tag{6.10}$$

6.2 Final Flow Properties at Each Station

Table 6.1 summarizes the Mach number, static pressure, and static temperature at all major engine stations.

Table 6.1: Flow properties at major engine stations

Station	Mach	Pressure (Pa)	Temperature (K)
1 (Freestream)	2.750	12110.00	216.50
4 (Intake Exit)	0.725	173518.44	492.21
5 (Diffuser Exit)	0.200	239435.49	539.64
6 (Combustor Exit)	0.507	185876.43	2092.31
e (Nozzle Exit)	2.511	12110.00	973.00

6.3 Combustor Performance Summary

The combustion process increases the total temperature from $T_{0,5} = 543.96$ K to:

$$T_{0,6} = 2200 \text{ K} \tag{6.11}$$

The heat added per unit mass is:

$$q = 1663.50 \text{ kJ/kg} \tag{6.12}$$

After accounting for Rayleigh heat addition and a 5% additional pressure loss, the combustor exit total pressure is:

$$P_{0,6} = 210491.63 \text{ Pa} \tag{6.13}$$

6.4 Overall Engine Performance

The nozzle expands the combustor exit flow to ambient pressure, producing the final engine thrust. The key performance parameters are:

$$\dot{m} = 9.880 \text{ kg/s} \tag{6.14}$$

$$V_e = 1570.05 \text{ m/s} \tag{6.15}$$

$$F = 7498.45 \text{ N} \tag{6.16}$$

Thus, at the design flight condition ($M_1 = 2.75$), the ramjet engine produces approximately **7.5 kN of thrust**.

6.5 MATLAB Final Output

Figures 6.1 and 6.2 show the MATLAB outputs summarizing the final engine results.

```
================= INTAKE RESULTS =================
Theta1 = 14.30 deg   Theta2 = 17.40 deg
Beta1  = 33.58 deg   Beta2  = 46.31 deg
Wedge Length L1 = 0.3766 m
Wedge Length L2 = 0.2388 m
Total Pressure Recovery P04/P01 = 0.808695

================= AREAS =================
A4 (Intake exit / Diffuser inlet)   = 0.022720 m^2
A5 (Diffuser exit / Combustor inlet)= 0.062500 m^2
A6 (Combustor exit)                 = 0.062500 m^2
At (Nozzle throat)                  = 0.054469 m^2
Ae (Nozzle exit)                    = 0.145106 m^2
Nozzle Area Ratio Ae/At             = 2.6640

================= COMBUSTOR =================
Heat Added q = 1663.50 kJ/kg
T06 = 2200.00 K
P06 = 210491.63 Pa
M6  = 0.5073
```

Figure 6.1: MATLAB summary of intake, geometry, and combustor results.

```
================= STATION TABLE =================
Station |   Mach   |    P (Pa)    |    T (K)
-------------------------------------------------
1  (Freestream)      2.750   12110.00   216.50
4  (Intake Exit)     0.725   173518.44   492.21
5  (Diffuser Exit)   0.200   239435.49   539.64
6  (Combustor Exit)  0.507   185876.43   2092.31
e  (Nozzle Exit)     2.511   12110.00   973.00

================= FINAL PERFORMANCE =================
Mass Flow Rate = 9.880 kg/s
Exit Velocity  = 1570.05 m/s
Net Thrust     = 7498.45 N
=====================================================
```

Figure 6.2: MATLAB station table and final engine performance results.

6.6 Discussion

The optimized two–ramp intake achieved a total pressure recovery of approximately 81%, providing efficient compression of the freestream flow. The diffuser further decelerated the flow to a low Mach number appropriate for stable combustion. Heat addition in the combustor raised the gas temperature to 2200 K, and the convergent–divergent nozzle successfully expanded the flow to ambient pressure. The resulting engine performance meets the project design objectives.

Chapter 7

Bonus: Off–Design Performance Analysis

7.1 Objective

In practical operation, a ramjet engine rarely flies exactly at its design Mach number. Therefore, a bonus off–design analysis was performed to evaluate engine performance when the flight Mach number deviates from the design point, while keeping the engine geometry fixed.

The intake ramp angles, duct areas, and nozzle geometry obtained from the design–point optimization were held constant. The inlet Mach number M_1 was then varied, and the full engine flow was recomputed using the same gas–dynamic methodology as in the design analysis.

Two off–design cases were investigated:

- Case A: $M_1 = 2$
- Case B: $M_1 = 4$

This analysis provides insight into how sensitive the ramjet performance is to changes in flight speed.

7.1.1 Fixed Geometry Used in Bonus Study

The following geometry obtained from the design–point optimization was held fixed:

$$\theta_1 = 14.30^\circ, \quad \theta_2 = 17.40^\circ \tag{7.1}$$

$$A_4 = 0.022720 \text{ m}^2 \tag{7.2}$$

$$A_5 = 0.062500 \text{ m}^2 \tag{7.3}$$

$$A_6 = 0.062500 \text{ m}^2 \tag{7.4}$$

$$A_t = 0.054469 \text{ m}^2 \tag{7.5}$$

$$A_e = 0.145106 \text{ m}^2 \tag{7.6}$$

$$\frac{A_e}{A_t} = 2.664 \tag{7.7}$$

The combustor maximum total temperature was again limited to:

$$T_{0,6} = 2200 \text{ K} \tag{7.8}$$

7.1.2 Case A: Inlet Mach Number $M_1 = 2$

For the off–design case with inlet Mach number $M_1 = 2$, it was observed that the fixed intake ramp deflection angles obtained from the design point ($\theta_1 = 14.3°$, $\theta_2 = 17.4°$) exceed the maximum allowable deflection angle for attached oblique shocks at this lower Mach number. Consequently, the shock waves would become detached in front of the intake, leading to a bow shock configuration rather than the attached shock system assumed in the present intake model.

Modeling detached shock behavior requires solving two–dimensional shock–detachment flow fields, which is beyond the scope of the one–dimensional gas–dynamics methods covered in this course. Therefore, the $M_1 = 2$ case is included only as a qualitative off–design demonstration, and no further intake optimization for attached shocks was pursued at this Mach number.

7.1.3 Case B: Inlet Mach Number $M_1 = 4$

(MatLab Code found at Appendix 9.6)

In the second off–design case, the freestream Mach number was increased to $M_1 = 4$. At this higher Mach number, the fixed intake geometry generates stronger shock losses, modifying the total pressure recovery. The downstream diffuser, combustor, and nozzle flow were again recomputed under the same assumptions.

The resulting engine performance for this case is shown in Figure 7.1.

```
================ BONUS OFF-DESIGN (M1 = 4.00) ================

--- Fixed Geometry ---
theta1=14.30 deg, theta2=17.40 deg
A4=0.022720, A5=0.062500, A6=0.062500, At=0.054469, Ae=0.145106 (m^2)

--- Intake (recomputed) ---
beta1=26.352 deg, beta2=34.960 deg
M2=2.9793, M3=2.1170, M4=0.5587
P04/P01 = 0.467886
Wedge lengths: L1=0.5047 m, L2=0.3576 m

--- Diffuser (fixed A5/A4) ---
A5/A4 = 2.7509 -> solved M5 = 0.1724

--- Combustor ---
T06 = 2200.00 K, P06 = 792045.77 Pa, M6 = 0.2853
q_add = 1296.51 kJ/kg

--- Nozzle ---
Ae/At = 2.6640 -> Me = 2.5110
Pe (isentropic) = 45568.43 Pa, ambient = 12110.00 Pa
Nozzle regime: Underexpanded (Pe > Pambient)
Te = 973.00 K, Ve = 1570.05 m/s

--- Performance ---
mdot = 14.371 kg/s
Thrust = 10463.70 N
============================================================
```

Figure 7.1: MATLAB output for off–design case $M_1 = 4$ with fixed engine geometry.

7.1.4 Discussion of Off–Design Behavior

The off–design results show that:

- At lower inlet Mach numbers, shock losses are reduced, but the mass flow rate and exit velocity decrease, leading to lower thrust.
- At higher inlet Mach numbers, stronger shock waves produce larger total pressure losses in the intake, but the higher captured mass flow and exit velocity result in increased thrust.
- The nozzle becomes underexpanded in the high–Mach case, indicating that the fixed nozzle geometry is no longer ideally matched to ambient pressure.

These results confirm that ramjet engines are highly sensitive to flight Mach number and that fixed–geometry designs achieve peak efficiency only near their design point.

7.1.5 Summary

The bonus off–design analysis demonstrates the performance variation of the optimized ramjet when operating away from the design Mach number. This provides a realistic assessment of engine behavior in variable flight conditions and highlights the importance of intake–nozzle matching for stable and efficient operation.

Chapter 8

Conclusion

The ramjet engine was successfully designed using gas-dynamic principles. Optimizing intake shock structure significantly improved pressure recovery and overall thrust. The integrated analysis demonstrates the strong coupling between intake performance, combustor heat addition, and nozzle expansion in ramjet engines.

Chapter 9

Appendix: MATLAB Codes

9.1 MATLAB Code for Intake Shock Optimization

This appendix includes the MATLAB code used to optimize the ramjet intake geometry at a design Mach number of $M_1 = 2.75$. The script evaluates a two–oblique-shock intake followed by a terminal normal shock and identifies the wedge deflection angles that maximize the total pressure recovery $P_{0,4}/P_{0,1}$. A contour plot is also generated to visualize the variation of pressure recovery with intake geometry.

```
%% ==================Intake============================
clc; clear; close all;

gamma = 1.4;
M1    = 2.75;

theta1_range = 0.5 : 0.1 : 25;
theta2_range = 0.5 : 0.1 : 25;

opts_fzero = optimset('Display','off');

best = struct('theta1',NaN,'theta2',NaN,'rec',-Inf,...
              'beta1',NaN,'beta2',NaN,'M2',NaN,'M3',NaN,'M4',NaN);

count_feasible = 0;

for theta1 = theta1_range
    [ok1, beta1, M2, P02_P01] = obliqueStage(M1, theta1, gamma,
       opts_fzero);
    if ~ok1, continue; end
    for theta2 = theta2_range
        [ok2, beta2, M3, P03_P02] = obliqueStage(M2, theta2, gamma,
           opts_fzero);
        if ~ok2 || M3 <= 1, continue; end
        P04_P03 = totalPressureRatioNormalShock(M3, gamma);
        M4      = downstreamMachNormalShock(M3, gamma);
        recovery = P02_P01 * P03_P02 * P04_P03;
        count_feasible = count_feasible + 1;
        if recovery > best.rec
            best.theta1 = theta1;
            best.theta2 = theta2;
            best.rec    = recovery;
            best.beta1  = beta1;
```

```
            best.beta2  = beta2;
            best.M2     = M2;
            best.M3     = M3;
            best.M4     = M4;
        end
    end
end
%% ----------------------------------------
fprintf('================ OPTIMUM INTAKE DESIGN ================\n');
fprintf('Feasible designs checked: %d\n\n', count_feasible);

fprintf('Best theta1   = %.3f deg\n', best.theta1);
fprintf('Best theta2   = %.3f deg\n', best.theta2);
fprintf('Best beta1    = %.3f deg\n', best.beta1);
fprintf('Best beta2    = %.3f deg\n', best.beta2);
fprintf('Mach after shock 1 (M2) = %.4f\n', best.M2);
fprintf('Mach after shock 2 (M3) = %.4f\n', best.M3);
fprintf('Mach after normal shock (M4) = %.4f\n', best.M4);
fprintf('Total pressure recovery (P0,4/P0,1) = %.6f\n', best.rec);
fprintf('=======================================================\n');

%% ----------------------------------------
REC = nan(length(theta1_range), length(theta2_range));
for i = 1:length(theta1_range)
    th1 = theta1_range(i);
    [ok1, ~, M2, P02_P01] = obliqueStage(M1, th1, gamma, opts_fzero);
    if ~ok1, continue; end
    for j = 1:length(theta2_range)
        th2 = theta2_range(j);
        [ok2, ~, M3, P03_P02] = obliqueStage(M2, th2, gamma, opts_fzero
            );
        if ~ok2 || M3 <= 1, continue; end
        P04_P03 = totalPressureRatioNormalShock(M3, gamma);
        REC(i,j) = P02_P01 * P03_P02 * P04_P03;
    end
end

figure;
contourf(theta2_range, theta1_range, REC, 30, 'LineColor','none');
colorbar; grid on;
xlabel('\theta_2 (deg)'); ylabel('\theta_1 (deg)');
title('Total Pressure Recovery P0,4 / P0,1');
hold on;
plot(best.theta2, best.theta1, 'wo', 'MarkerFaceColor','w', 'MarkerSize
    ',7);
text(best.theta2, best.theta1, '  Optimum', 'Color','w', 'FontWeight','
    bold');

%% ========================FUNCTIONS=======================

function [ok, beta_deg, M2, P0out_P0in] = obliqueStage(M1, theta_deg,
    gamma, opts_fzero)
    ok = false; beta_deg = NaN; M2 = NaN; P0out_P0in = NaN;
    theta = deg2rad(theta_deg);
    theta_max = thetaMaxAttached(M1, gamma);
    if theta >= theta_max
        return;
    end
```

```
86
87      beta_min = asin(1/M1) + 1e-6;
88      beta_max = pi/2 - 1e-6;
89      f = @(beta) tan(theta) - 2*cot(beta) .* ...
90          ((M1^2*sin(beta).^2 - 1) ./ (M1^2*(gamma + cos(2*beta)) + 2));
91      beta_guess = (beta_min + beta_max)/2;
92
93      try
94          beta = fzero(f, beta_guess, opts_fzero);
95      catch
96          return;
97      end
98
99      if ~(beta > beta_min && beta < beta_max)
100         return;
101     end
102
103     Mn1 = M1*sin(beta);
104     P0out_P0in = totalPressureRatioNormalShock(Mn1, gamma);
105     Mn2 = sqrt((1 + 0.5*(gamma-1)*Mn1^2) / (gamma*Mn1^2 - 0.5*(gamma-1)
            ));
106     M2 = Mn2 / sin(beta - theta);
107     beta_deg = rad2deg(beta);
108     ok = (M2 > 0);
109 end
110
111 function theta_max = thetaMaxAttached(M, gamma)
112     beta_min = asin(1/M) + 1e-6;
113     beta_max = pi/2 - 1e-6;
114     betas = linspace(beta_min, beta_max, 5000);
115     thetas = atan( 2*cot(betas) .* ...
116         ((M^2*sin(betas).^2 - 1) ./ (M^2*(gamma + cos(2*betas)) + 2)) )
                ;
117     theta_max = max(thetas);
118 end
119
120 function P02_P01 = totalPressureRatioNormalShock(M1, gamma)
121     p2_p1 = 1 + 2*gamma/(gamma+1) * (M1^2 - 1);
122     rho2_rho1 = ((gamma+1)*M1^2) / ((gamma-1)*M1^2 + 2);
123     T2_T1 = p2_p1 / rho2_rho1;
124     M2 = downstreamMachNormalShock(M1, gamma);
125     P01 = 1 * (1 + 0.5*(gamma-1)*M1^2)^(gamma/(gamma-1)); % normalized
            by p1=1
126     P02 = p2_p1 * (1 + 0.5*(gamma-1)*M2^2)^(gamma/(gamma-1)); %
            normalized by p1=1
127     P02_P01 = P02 / P01;
128     if P02_P01 > 1
129         P02_P01 = min(P02_P01, 1);
130     end
131 end
132
133 function M2 = downstreamMachNormalShock(M1, gamma)
134     M2 = sqrt((1 + 0.5*(gamma-1)*M1^2) / (gamma*M1^2 - 0.5*(gamma-1)));
135 end
```

Listing 9.1: MATLAB code for ramjet intake shock optimization

9.2 MATLAB Code for Diffuser Analysis

This MATLAB script evaluates the subsonic diffuser following the intake. It computes the static flow properties and the required area ratio for decelerating the flow from the intake exit Mach number to the combustor entry Mach number.

```
%% ==================Diffuser============================

clc; clear; close all;

gamma = 1.4;

%% -------------------
M4   = 0.7250;
P04_P01 = 0.808695;
P1 = 12.11e3;
T1 = 216.5;
M1 = 2.75;

%% --------------------
T01 = T1*(1 + (gamma-1)/2*M1^2);
P01 = P1*(1 + (gamma-1)/2*M1^2)^(gamma/(gamma-1));

%% --------------------
P04 = P04_P01 * P01;
T04 = T01;

%% ============================================================
M5 = 0.20;     % Typical subsonic diffuser exit Mach

%% --------------------
T4 = T04 / (1 + (gamma-1)/2*M4^2);
P4 = P04 / (1 + (gamma-1)/2*M4^2)^(gamma/(gamma-1));

%% -------------------
T5 = T04 / (1 + (gamma-1)/2*M5^2);
P5 = P04 / (1 + (gamma-1)/2*M5^2)^(gamma/(gamma-1));

%% --------------------
A5_A4 = (M4/M5)* ...
    ((1 + (gamma-1)/2*M5^2)/(1 + (gamma-1)/2*M4^2))^((gamma+1)/(2*(
       gamma-1)));

%% ============================================================
fprintf('================ DIFFUSER RESULTS ================\n');
fprintf('Inlet Mach (after intake)       M4 = %.4f\n',M4);
fprintf('Exit Mach (combustor entry)     M5 = %.4f\n\n',M5);
fprintf('Total Pressure (constant)       P04 = %.2f Pa\n',P04);
fprintf('Total Temperature (constant)    T04 = %.2f K\n\n',T04);
fprintf('Static Pressure at diffuser inlet   P4 = %.2f Pa\n',P4);
fprintf('Static Pressure at diffuser exit    P5 = %.2f Pa\n\n',P5);
fprintf('Static Temperature at inlet     T4 = %.2f K\n',T4);
fprintf('Static Temperature at exit      T5 = %.2f K\n\n',T5);
fprintf('Required Area Ratio A5/A4 = %.4f\n',A5_A4);
fprintf('==================================================\n');
```

Listing 9.2: MATLAB code for diffuser analysis

9.3 MATLAB Code for Combustor Analysis

This script models heat addition in the combustor using Rayleigh flow relations. It evaluates choking, exit conditions, and the heat added per unit mass under a maximum total temperature constraint.

```
%% ==================Combustor============================
clc; clear; close all;

%% ---------------------------------------
gamma = 1.4;
R     = 287;
cp    = gamma*R/(gamma-1);
M5  = 0.2;
P05 = 246206.86;
T05 = 543.96;
P5  = 239435.36;
T5  = 539.64;
T0_max = 2200;
P0_extra_loss = 0.95;

%% ---------------------------------------
T0_T0star_5 = rayleigh_T0_T0star(M5, gamma);
T0star = T05 / T0_T0star_5;
T06_desired = min(T0_max, T0star);
if T0_max > T0star
    choked = true;
else
    choked = false;
end
target_T0_T0star = T06_desired / T0star;

%% ---------------------------------------
if choked
    M6 = 0.999;
else
    f = @(M) rayleigh_T0_T0star(M, gamma) - target_T0_T0star;
    M_low  = max(1e-4, M5);
    M_high = 0.999;
    if f(M_low)*f(M_high) < 0
        M6 = fzero(f, [M_low, M_high]);
    else
        M6 = fminbnd(@(M) abs(f(M)), M_low, M_high);
    end
end

%% ---------------------------------------
p_pstar_5  = rayleigh_p_pstar(M5, gamma);
T_Tstar_5  = rayleigh_T_Tstar(M5, gamma);
p0_p0star_5 = rayleigh_p0_p0star(M5, gamma);
p_pstar_6  = rayleigh_p_pstar(M6, gamma);
T_Tstar_6  = rayleigh_T_Tstar(M6, gamma);
p0_p0star_6 = rayleigh_p0_p0star(M6, gamma);
pstar = P5 / p_pstar_5;
Tstar = T5 / T_Tstar_5;
P6 = p_pstar_6 * pstar;
T6 = T_Tstar_6 * Tstar;
T06 = T06_desired;
```

```
P06_rayleigh = P05 * (p0_p0star_6 / p0_p0star_5);
P06 = P0_extra_loss * P06_rayleigh;

%% ----------------------------------------
q_add = cp * (T06 - T05);     % J/kg

%% ----------------------------------------
fprintf('================ COMBUSTOR RESULTS (RAYLEIGH) ================\n');
fprintf('Combustor entry (station 5):\n');
fprintf('  M5   = %.4f\n  T05  = %.2f K (total)\n  P05  = %.2f Pa (total)\n', M5, T05, P05);
fprintf('  T5   = %.2f K (static)\n  P5   = %.2f Pa (static)\n\n', T5, P5);
fprintf('Combustor exit (station 6):\n');
fprintf('  M6   = %.4f\n', M6);
fprintf('  T06  = %.2f K (total)   [limit = %.2f K]\n', T06, T0_max);
fprintf('  P06 (Rayleigh only) = %.2f Pa (total)\n', P06_rayleigh);
fprintf('  P06 (with extra 5%% loss) = %.2f Pa (total)\n', P06);
fprintf('  T6   = %.2f K (static)\n  P6   = %.2f Pa (static)\n\n', T6, P6);
fprintf('Heat added per unit mass:\n');
fprintf('  q = cp*(T06 - T05) = %.2f kJ/kg\n', q_add/1000);
if choked
    fprintf('\n*** NOTE: Desired T0_max exceeds Rayleigh choking limit for this inlet.\n');
    fprintf('*** Flow is CHOKED in combustor (M -> 1). T06 capped at T0* = %.2f K.\n', T0star);
end
fprintf('============================================================\n');

%% ====================== FUNCTIONS ==========================

function val = rayleigh_p_pstar(M, g)
    val = (g + 1) ./ (1 + g*M.^2);
end

function val = rayleigh_T_Tstar(M, g)
    val = ((g + 1).^2 .* M.^2) ./ (1 + g*M.^2).^2;
end

function val = rayleigh_T0_T0star(M, g)
    T_Ts = rayleigh_T_Tstar(M, g);
    val  = T_Ts .* (1 + (g-1)/2 .* M.^2) ./ (1 + (g-1)/2);
end

function val = rayleigh_p0_p0star(M, g)
    p_ps = rayleigh_p_pstar(M, g);
    val  = p_ps .* ((1 + (g-1)/2 .* M.^2).^(g/(g-1))) ./ ((1 + (g-1)/2).^(g/(g-1)));
end
```

Listing 9.3: MATLAB code for Rayleigh-flow combustor analysis

9.4 MATLAB Code for Nozzle Analysis

This script computes the nozzle exit conditions assuming isentropic expansion to ambient pressure and evaluates the resulting mass flow rate and thrust.

```
%% ==================Nozzle============================
clc; clear; close all;

gamma = 1.4;
R = 287;

%% --------------------------------
T06 = 2200.00;
P06 = 210491.78;

%% --------------------------------
Pa = 12.11e3;
Ta = 216.5;

%% ------------------------------------
side_inlet = 0.25;
A_inlet = side_inlet^2;

%% ================================================================
f = @(M) P06 ./ (1 + (gamma-1)/2 .* M.^2).^(gamma/(gamma-1)) - Pa;
Me = fzero(f,[1.01 8]);

%% ================================================================
Te = T06 / (1 + (gamma-1)/2 * Me^2);
ae = sqrt(gamma*R*Te);
Ve = Me * ae;

%% ================================================================
A_ratio = @(M) (1./M).* ...
    ((2/(gamma+1))*(1+(gamma-1)/2*M.^2)).^((gamma+1)/(2*(gamma-1)));
Ae_At = A_ratio(Me);

%% ================================================================
M1 = 2.75;
T1 = Ta;
P1 = Pa;
rho1 = P1/(R*T1);
a1 = sqrt(gamma*R*T1);
V1 = M1*a1;
mdot = rho1 * V1 * A_inlet;

%% ================================================================
Thrust = mdot*(Ve - V1);

%% ================================================================
fprintf('================ NOZZLE RESULTS ================\n');
fprintf('Exit Mach number Me       = %.4f\n',Me);
fprintf('Exit static pressure Pe   = %.2f Pa\n',Pa);
fprintf('Exit static temperature   = %.2f K\n',Te);
fprintf('Exit velocity Ve          = %.2f m/s\n\n',Ve);

fprintf('Nozzle area ratio Ae/At   = %.4f\n',Ae_At);

```

```
fprintf('Mass flow rate m_dot      = %.3f kg/s\n',mdot);
fprintf('Net Thrust                = %.2f N\n',Thrust);
fprintf('=================================================\n');
```

Listing 9.4: MATLAB code for isentropic nozzle analysis

9.5 MATLAB Code for Complete Ramjet Engine Model

This script integrates the intake, diffuser, combustor, and nozzle models to evaluate overall engine performance and net thrust.

```
%% ==================Full Project============================
clc; clear; close all;

gamma = 1.4;
R = 287;
cp = gamma*R/(gamma-1);

%% ==================================
M1 = 2.75;
P1 = 12.11e3;
T1 = 216.5;

%% ======================================
side_duct = 0.25;
A_duct = side_duct^2;

%% ================================================
theta1_range = 0.5:0.1:25;
theta2_range = 0.5:0.1:25;
opts = optimset('Display','off');
best.rec = -Inf;
for th1 = theta1_range
    [ok1,beta1,M2,P02_P01] = obliqueStage(M1,th1,gamma,opts);
    if ~ok1, continue; end
    for th2 = theta2_range
        [ok2,beta2,M3,P03_P02] = obliqueStage(M2,th2,gamma,opts);
        if ~ok2 || M3<=1, continue; end
        P04_P03 = totalPressureRatioNormalShock(M3,gamma);
        M4 = downstreamMachNormalShock(M3,gamma);
        rec = P02_P01 * P03_P02 * P04_P03;
        if rec > best.rec
            best = struct('theta1',th1,'theta2',th2,...
                          'beta1',beta1,'beta2',beta2,...
                          'M2',M2,'M3',M3,'M4',M4,'rec',rec);
        end
    end
end

%% --------
T01 = T1*(1+(gamma-1)/2*M1^2);
P01 = P1*(1+(gamma-1)/2*M1^2)^(gamma/(gamma-1));
T04 = T01;
P04 = best.rec * P01;
M4  = best.M4;
T4 = T04/(1+(gamma-1)/2*M4^2);
```

```
P4 = P04/(1+(gamma-1)/2*M4^2)^(gamma/(gamma-1));

%% ================================================
M5 = 0.20;
T5 = T04/(1+(gamma-1)/2*M5^2);
P5 = P04/(1+(gamma-1)/2*M5^2)^(gamma/(gamma-1));
A5_A4 = (M4/M5)*((1+(gamma-1)/2*M5^2)/(1+(gamma-1)/2*M4^2))^((gamma+1)
    /(2*(gamma-1)));
A5 = A_duct;
A6 = A5;
A4 = A5 / A5_A4;
A_capture = A5;
T05 = T04;
P05 = P04;

%% ================================================
T0_max = 2200;
P0_extra_loss = 0.95;
T0_T0star_5 = rayleigh_T0_T0star(M5,gamma);
T0star = T05/T0_T0star_5;
T06 = min(T0_max,T0star);
f = @(M) rayleigh_T0_T0star(M,gamma) - (T06/T0star);
M6 = fzero(f,[M5 0.999]);
p_pstar_5   = rayleigh_p_pstar(M5,gamma);
T_Tstar_5   = rayleigh_T_Tstar(M5,gamma);
p0_p0star_5 = rayleigh_p0_p0star(M5,gamma);
p_pstar_6   = rayleigh_p_pstar(M6,gamma);
T_Tstar_6   = rayleigh_T_Tstar(M6,gamma);
p0_p0star_6 = rayleigh_p0_p0star(M6,gamma);
pstar = P5/p_pstar_5;
Tstar = T5/T_Tstar_5;
P6 = p_pstar_6*pstar;
T6 = T_Tstar_6*Tstar;
P06_rayleigh = P05*(p0_p0star_6/p0_p0star_5);
P06 = P0_extra_loss * P06_rayleigh;
q_add = cp*(T06-T05);   % J/kg

%% ================================================
Pa = P1;
f = @(M) P06/(1+(gamma-1)/2*M.^2).^(gamma/(gamma-1)) - Pa;
Me = fzero(f,[1.01 8]);
Te = T06/(1+(gamma-1)/2*Me^2);
ae = sqrt(gamma*R*Te);
Ve = Me*ae;
A_ratio = @(M) (1./M).* ...
 ((2/(gamma+1))*(1+(gamma-1)/2*M.^2)).^((gamma+1)/(2*(gamma-1)));
Ae_At = A_ratio(Me);
rho1 = P1/(R*T1);
a1 = sqrt(gamma*R*T1);
V1 = M1*a1;
mdot = rho1 * V1 * A_capture;
mdot_per_At = (P06/sqrt(T06))*sqrt(gamma/R)* ...
              ((gamma+1)/2)^(-(gamma+1)/(2*(gamma-1)));
At = mdot / mdot_per_At;
Ae = Ae_At * At;
Thrust = mdot*(Ve - V1);

%% ================================================
```

```
h = sqrt(A_capture);
beta1 = deg2rad(best.beta1);
beta2 = deg2rad(best.beta2);
L1 = h / tan(beta1);
L2 = h / tan(beta2);

%% ================================================
fprintf('\n================ INTAKE RESULTS ================\n');
fprintf('Theta1 = %.2f deg    Theta2 = %.2f deg\n',best.theta1,best.
    theta2);
fprintf('Beta1   = %.2f deg    Beta2   = %.2f deg\n',best.beta1,best.beta2
    );
fprintf('Wedge Length L1 = %.4f m\n',L1);
fprintf('Wedge Length L2 = %.4f m\n',L2);
fprintf('Total Pressure Recovery P04/P01 = %.6f\n',best.rec);

fprintf('\n================ AREAS ================\n');
fprintf('A4 (Intake exit / Diffuser inlet)    = %.6f m^2\n',A4);
fprintf('A5 (Diffuser exit / Combustor inlet) = %.6f m^2\n',A5);
fprintf('A6 (Combustor exit)                  = %.6f m^2\n',A6);
fprintf('At (Nozzle throat)                   = %.6f m^2\n',At);
fprintf('Ae (Nozzle exit)                     = %.6f m^2\n',Ae);
fprintf('Nozzle Area Ratio Ae/At              = %.4f\n',Ae_At);

fprintf('\n================ COMBUSTOR ================\n');
fprintf('Heat Added q = %.2f kJ/kg\n',q_add/1000);
fprintf('T06 = %.2f K\n',T06);
fprintf('P06 = %.2f Pa\n',P06);
fprintf('M6  = %.4f\n',M6);

fprintf('\n================ STATION TABLE ================\n');
fprintf('Station | Mach | P (Pa) | T (K)\n');
fprintf('-----------------------------------------------\n');
fprintf('1  (Freestream)  %.3f  %.2f  %.2f\n',M1,P1,T1);
fprintf('4  (Intake Exit) %.3f  %.2f  %.2f\n',M4,P4,T4);
fprintf('5  (Diffuser Exit) %.3f  %.2f  %.2f\n',M5,P5,T5);
fprintf('6  (Combustor Exit) %.3f  %.2f  %.2f\n',M6,P6,T6);
fprintf('e  (Nozzle Exit) %.3f  %.2f  %.2f\n',Me,Pa,Te);

fprintf('\n================ FINAL PERFORMANCE ================\n');
fprintf('Mass Flow Rate = %.3f kg/s\n',mdot);
fprintf('Exit Velocity  = %.2f m/s\n',Ve);
fprintf('Net Thrust     = %.2f N\n',Thrust);
fprintf('================================================\n');

%% ==================== FUNCTIONS ==============================
function [ok,beta_deg,M2,P0out_P0in] = obliqueStage(M1,theta_deg,gamma,
    opts)
ok=false; beta_deg=NaN; M2=NaN; P0out_P0in=NaN;
theta = deg2rad(theta_deg);
theta_max = thetaMaxAttached(M1,gamma);
if theta>=theta_max, return; end
beta_min=asin(1/M1)+1e-6; beta_max=pi/2-1e-6;
f=@(b) tan(theta)-2*cot(b).*((M1^2*sin(b).^2-1)./(M1^2*(gamma+cos(2*b))
    +2));
try beta=fzero(f,(beta_min+beta_max)/2,opts); catch; return; end
Mn1=M1*sin(beta);
P0out_P0in=totalPressureRatioNormalShock(Mn1,gamma);
```

```
Mn2=sqrt((1+0.5*(gamma-1)*Mn1^2)/(gamma*Mn1^2-0.5*(gamma-1)));
M2=Mn2/sin(beta-theta);
beta_deg=rad2deg(beta);
ok=true;
end

function theta_max=thetaMaxAttached(M,gamma)
bmin=asin(1/M)+1e-6; bmax=pi/2-1e-6;
b=linspace(bmin,bmax,5000);
theta=atan(2*cot(b).*((M^2*sin(b).^2-1)./(M^2*(gamma+cos(2*b))+2)));
theta_max=max(theta);
end

function P02_P01=totalPressureRatioNormalShock(M1,gamma)
p2_p1=1+2*gamma/(gamma+1)*(M1^2-1);
M2=downstreamMachNormalShock(M1,gamma);
P01=(1+0.5*(gamma-1)*M1^2)^(gamma/(gamma-1));
P02=p2_p1*(1+0.5*(gamma-1)*M2^2)^(gamma/(gamma-1));
P02_P01=P02/P01;
end

function M2=downstreamMachNormalShock(M1,gamma)
M2=sqrt((1+0.5*(gamma-1)*M1^2)/(gamma*M1^2-0.5*(gamma-1)));
end

function val=rayleigh_p_pstar(M,g)
val=(g+1)./(1+g*M.^2);
end

function val=rayleigh_T_Tstar(M,g)
val=((g+1).^2.*M.^2)./(1+g*M.^2).^2;
end

function val=rayleigh_T0_T0star(M,g)
T_Ts=rayleigh_T_Tstar(M,g);
val=T_Ts.*(1+(g-1)/2.*M.^2)./(1+(g-1)/2);
end

function val=rayleigh_p0_p0star(M,g)
p_ps=rayleigh_p_pstar(M,g);
val=p_ps.*((1+(g-1)/2.*M.^2).^(g/(g-1)))./((1+(g-1)/2).^(g/(g-1)));
end
```

Listing 9.5: MATLAB code for full ramjet engine performance model

9.6 MATLAB Code for the Bonus Part M=4

```
%% ==================Bonus M=4============================
clc; clear; close all;

gamma = 1.4;
R = 287;
cp = gamma*R/(gamma-1);

%% ==================================
M1 = 4.0;
```

```
P1 = 12.11e3;
T1 = 216.5;

%% ================================
theta1 = 14.30;
theta2 = 17.40;
A4 = 0.022720;
A5 = 0.062500;
A6 = 0.062500;
At = 0.054469;
Ae = 0.145106;
A_capture = A5;

%% ================================
T01 = T1*(1 + (gamma-1)/2*M1^2);
P01 = P1*(1 + (gamma-1)/2*M1^2)^(gamma/(gamma-1));

%% ================================================================
opts = optimset('Display','off');
[ok1, beta1, M2, P02_P01] = obliqueStage(M1, theta1, gamma, opts);
if ~ok1
    error('Shock 1 detached at M1=%.2f, theta1=%.2f deg.', M1, theta1);
end

[ok2, beta2, M3, P03_P02] = obliqueStage(M2, theta2, gamma, opts);
if ~ok2
    error('Shock 2 detached at M2=%.3f, theta2=%.2f deg.', M2, theta2);
end

if M3 <= 1
    error('M3 <= 1 after oblique shocks (M3=%.3f). Terminal normal
       shock cannot form.', M3);
end

P04_P03 = totalPressureRatioNormalShock(M3, gamma);
M4 = downstreamMachNormalShock(M3, gamma);
P04_P01 = P02_P01 * P03_P02 * P04_P03;
T04 = T01;
P04 = P04_P01 * P01;
T4 = T04/(1 + (gamma-1)/2*M4^2);
P4 = P04/(1 + (gamma-1)/2*M4^2)^(gamma/(gamma-1));

%% ================================================================
A5_A4 = A5 / A4;
A_Astar = @(M) (1./M).* ...
    ((2/(gamma+1))*(1 + (gamma-1)/2*M.^2)).^((gamma+1)/(2*(gamma-1)));
A4_Astar = A_Astar(M4);
A5_Astar_target = A5_A4 * A4_Astar;
g = @(M) A_Astar(M) - A5_Astar_target;
M5 = fzero(g, [1e-4, 0.999]);
T05 = T04;
P05 = P04;
T5 = T05/(1 + (gamma-1)/2*M5^2);
P5 = P05/(1 + (gamma-1)/2*M5^2)^(gamma/(gamma-1));

%% ================================================================
T0_max = 2200;
P0_extra_loss = 0.95;
```

```
T0_T0star_5 = rayleigh_T0_T0star(M5, gamma);
T0star = T05 / T0_T0star_5;
T06 = min(T0_max, T0star);
target = T06 / T0star;
fM = @(M) rayleigh_T0_T0star(M, gamma) - target;

if abs(T06 - T0star) < 1e-8
    M6 = 0.999; % choked
else
    M6 = fzero(fM, [M5, 0.999]);
end

p_pstar_5   = rayleigh_p_pstar(M5, gamma);
T_Tstar_5   = rayleigh_T_Tstar(M5, gamma);
p0_p0star_5 = rayleigh_p0_p0star(M5, gamma);
p_pstar_6   = rayleigh_p_pstar(M6, gamma);
T_Tstar_6   = rayleigh_T_Tstar(M6, gamma);
p0_p0star_6 = rayleigh_p0_p0star(M6, gamma);
pstar = P5 / p_pstar_5;
Tstar = T5 / T_Tstar_5;
P6 = p_pstar_6 * pstar;
T6 = T_Tstar_6 * Tstar;
P06_rayleigh = P05 * (p0_p0star_6 / p0_p0star_5);
P06 = P0_extra_loss * P06_rayleigh;
q_add = cp*(T06 - T05);

%% ============================================================
Ae_At = Ae / At;
h = @(M) A_Astar(M) - Ae_At;
Me = fzero(h, [1.01, 10]);   % supersonic branch
Te = T06/(1 + (gamma-1)/2*Me^2);
ae = sqrt(gamma*R*Te);
Ve = Me*ae;
Pe = P06/(1 + (gamma-1)/2*Me^2)^(gamma/(gamma-1));  % isentropic to
    exit area

if Pe > P1
    nozzle_regime = "Underexpanded (Pe > Pambient)";
    Pe_for_thrust = Pe;        % keep pressure thrust
elseif Pe < P1
    nozzle_regime = "Overexpanded (Pe < Pambient) -> assume separated";
    Pe_for_thrust = P1;        % separation guard: pressure thrust ~ 0
else
    nozzle_regime = "Perfectly expanded (Pe = Pambient)";
    Pe_for_thrust = Pe;
end

%% ============================================================
rho1 = P1/(R*T1);
a1 = sqrt(gamma*R*T1);
V1 = M1*a1;
mdot = rho1 * V1 * A_capture;
Thrust = mdot*(Ve - V1) + (Pe_for_thrust - P1)*Ae;

%% ============================================================
hcap = sqrt(A_capture);
Lw1 = hcap / tan(deg2rad(beta1));
Lw2 = hcap / tan(deg2rad(beta2));
```

```
124
125 %% ============================================================
126 fprintf('\n================ BONUS OFF-DESIGN (M1 = %.2f) ================\n', M1);
127 fprintf('\n--- Fixed Geometry ---\n');
128 fprintf('theta1=%.2f deg, theta2=%.2f deg\n', theta1, theta2);
129 fprintf('A4=%.6f, A5=%.6f, A6=%.6f, At=%.6f, Ae=%.6f (m^2)\n', A4,A5,A6,At,Ae);
130 fprintf('\n--- Intake (recomputed) ---\n');
131 fprintf('beta1=%.3f deg, beta2=%.3f deg\n', beta1, beta2);
132 fprintf('M2=%.4f, M3=%.4f, M4=%.4f\n', M2, M3, M4);
133 fprintf('P04/P01 = %.6f\n', P04_P01);
134 fprintf('Wedge lengths: L1=%.4f m, L2=%.4f m\n', Lw1, Lw2);
135 fprintf('\n--- Diffuser (fixed A5/A4) ---\n');
136 fprintf('A5/A4 = %.4f -> solved M5 = %.4f\n', A5_A4, M5);
137 fprintf('\n--- Combustor ---\n');
138 fprintf('T06 = %.2f K, P06 = %.2f Pa, M6 = %.4f\n', T06, P06, M6);
139 fprintf('q_add = %.2f kJ/kg\n', q_add/1000);
140 fprintf('\n--- Nozzle ---\n');
141 fprintf('Ae/At = %.4f -> Me = %.4f\n', Ae_At, Me);
142 fprintf('Pe (isentropic) = %.2f Pa, ambient = %.2f Pa\n', Pe, P1);
143 fprintf('Nozzle regime: %s\n', nozzle_regime);
144 fprintf('Te = %.2f K, Ve = %.2f m/s\n', Te, Ve);
145 fprintf('\n--- Performance ---\n');
146 fprintf('mdot = %.3f kg/s\n', mdot);
147 fprintf('Thrust = %.2f N\n', Thrust);
148 fprintf('============================================================\n');
149
150 %% ==========================Functions==================================
151 function [ok,beta_deg,M2,P0out_P0in] = obliqueStage(M1,theta_deg,gamma,opts)
152 ok=false; beta_deg=NaN; M2=NaN; P0out_P0in=NaN;
153 theta = deg2rad(theta_deg);
154 theta_max = thetaMaxAttached(M1,gamma);
155 if theta >= theta_max, return; end
156 beta_min = asin(1/M1) + 1e-6;
157 beta_max = pi/2 - 1e-6;
158 f = @(beta) tan(theta) - 2*cot(beta).* ...
159     ((M1^2*sin(beta).^2 - 1)./(M1^2*(gamma + cos(2*beta)) + 2));
160
161 try
162     beta = fzero(f, (beta_min+beta_max)/2, opts);
163 catch
164     return;
165 end
166
167 Mn1 = M1*sin(beta);
168 P0out_P0in = totalPressureRatioNormalShock(Mn1,gamma);
169 Mn2 = sqrt((1 + 0.5*(gamma-1)*Mn1^2)/(gamma*Mn1^2 - 0.5*(gamma-1)));
170 M2  = Mn2 / sin(beta - theta);
171 beta_deg = rad2deg(beta);
172 ok = true;
173 end
174
175 function theta_max=thetaMaxAttached(M,gamma)
176 bmin=asin(1/M)+1e-6; bmax=pi/2-1e-6;
```

```
b=linspace(bmin,bmax,5000);
theta=atan(2*cot(b).*((M^2*sin(b).^2-1)./(M^2*(gamma+cos(2*b))+2)));
theta_max=max(theta);
end

function P02_P01=totalPressureRatioNormalShock(M1,gamma)
p2_p1=1+2*gamma/(gamma+1)*(M1^2-1);
M2=downstreamMachNormalShock(M1,gamma);
P01=(1+0.5*(gamma-1)*M1^2)^(gamma/(gamma-1));
P02=p2_p1*(1+0.5*(gamma-1)*M2^2)^(gamma/(gamma-1));
P02_P01=P02/P01;
end

function M2=downstreamMachNormalShock(M1,gamma)
M2=sqrt((1+0.5*(gamma-1)*M1^2)/(gamma*M1^2-0.5*(gamma-1)));
end

function val=rayleigh_p_pstar(M,g)
val=(g+1)./(1+g*M.^2);
end

function val=rayleigh_T_Tstar(M,g)
val=((g+1).^2.*M.^2)./(1+g*M.^2).^2;
end

function val=rayleigh_T0_T0star(M,g)
T_Ts=rayleigh_T_Tstar(M,g);
val=T_Ts.*(1+(g-1)/2.*M.^2)./(1+(g-1)/2);
end

function val=rayleigh_p0_p0star(M,g)
p_ps=rayleigh_p_pstar(M,g);
val=p_ps.*((1+(g-1)/2.*M.^2).^(g/(g-1)))./((1+(g-1)/2).^(g/(g-1)));
end
```

Listing 9.6: MATLAB code for inlet Mach=4